

# Vagal Nerve Stimulation in Patients With Heart Failure and Reduced Ejection Fraction

## The ANTHEM-HFrEF Trial

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### ABSTRACT

**BACKGROUND** Vagal nerve stimulation (VNS) may reverse autonomic dysfunction in heart failure (HF). This paper describes results from a pivotal trial, while addressing challenges investigators face following premature trial termination.

**OBJECTIVES** We report VNS actions in patients with HF with reduced ejection fraction and investigators response to early discontinuation.

**METHODS** Randomization was 2:1 to VNS or control. Entry required NYHA functional class II or III, left ventricular ejection fraction (LVEF)  $\leq 35\%$ , and N-terminal pro-B-type natriuretic peptide  $\geq 800$  pg/mL. Primary efficacy endpoint was time to cardiovascular death or HF hospitalization. Secondary endpoints included LVEF, 6-minute walk distance, and Kansas City Cardiomyopathy Questionnaire. Adaptive sample sizing targeted  $\leq 1,000$  patients. *P* values are 1-sided.

**RESULTS** After randomizing 532 patients, the sponsor terminated enrollment, unrelated to futility or efficacy. The primary efficacy endpoint was not met (HR: 0.84; 95% CI: 0.62-1.12; *P* = 0.115). The primary safety endpoint achieved 96.7% freedom from procedure- or device-related serious adverse events. Autonomic engagement persisted long-term. Findings were favorable, although inconclusive, for the primary endpoint, HF hospitalization, Kansas City Cardiomyopathy Questionnaire, NYHA functional class, and 6-minute walk distance. LVEF was unchanged. The still-blinded investigators, seeking to extend data reporting while maintaining scientific integrity, had prespecified a single exploratory outcomes-symptoms win ratio. Results were favorable although merely hypothesis-generating, given primary endpoint neutrality, multiplicity, subjective bias, and marginal statistical significance.

**CONCLUSIONS** Early termination outside Data Monitoring Committee processes challenged investigators to maintain scientific and ethical integrity and transparency, while optimizing data reporting. The underpowered primary endpoint was neutral. Data documented feasibility, safety, and autonomic engagement. Trends favoring outcomes, symptoms, and function were inconclusive although hypothesis-generating. Importantly, we provide guidance for investigators facing trial discontinuation. (ANTHEM-HFrEF Pivotal Study [Autonomic Regulation Therapy to Enhance Myocardial Function and Reduce Progression of Heart Failure with Reduced Ejection Fraction]; [NCT03425422](https://doi.org/10.1016/j.jacc.2026.03.040)) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS  
AND ACRONYMS**6MWD** = 6-minute walk distance**CVD** = cardiovascular death**GDMT** = guideline-directed medical therapy**GEE** = generalized estimation equations**HFH** = heart failure hospitalization**HFREF** = heart failure with reduced ejection fraction**ITT** = intent to treat**KCCQ** = Kansas City Cardiomyopathy Questionnaire**LVEF** = left ventricular ejection fraction**NT-proBNP** = N-terminal pro-B-type natriuretic peptide**VNS** = vagal nerve stimulation

**A**NTHEM-HFrEF (Autonomic Regulation Therapy to Enhance Myocardial Function and Reduce Progression of Heart Failure with Reduced Ejection Fraction; [NCT03425422](#)) was a U.S. Food and Drug Administration (FDA)-informed, pivotal trial examining the efficacy and safety of vagal nerve stimulation (VNS) in patients with heart failure with reduced ejection fraction (HFrEF) who remain symptomatic despite guideline-directed medical therapy.<sup>1</sup> Shortly after the second interim analysis, during which the Data Monitoring Committee recommended continuing the trial unchanged, the sponsor discontinued enrollment and stopped the trial, in the absence of definitive signals for efficacy or safety concern. This action rendered the trial underpowered and changed the scientific and ethical responsibilities of the investigators to salvage as much scientific value from the trial as possible.

Correction of autonomic dysregulation through VNS<sup>2</sup> has previously been investigated, without a definitive conclusion,<sup>3,4</sup> perhaps related to the complexity of VNS dosing and variability in autonomic engagement.<sup>5</sup> The present pivotal randomized, controlled trial, informed by a prior pilot study,<sup>6</sup> incorporated adaptive sample sizing and the possibility of early device approval, based on symptom and function improvement. The significance of this report lies in both the trial's findings and the investigators' response to premature discontinuation, seeking to preserve scientific value and transparency under these circumstances.

## METHODS

**STUDY DESIGN.** ANTHEM-HFrEF was developed in collaboration with the FDA's Center for Devices and Radiological Health under the Expedited Access Pathway and Breakthrough Devices Programs.<sup>7</sup> The

study design was published previously.<sup>8</sup> A steering committee, clinical events committee (CEC), and data and safety monitoring committee provided study oversight. Patients were randomized 2:1 to VNS via VITARIA System right cervical vagus nerve implantation ([Central Illustration](#)) plus guideline-directed medical therapy (GDMT) or GDMT alone. Randomization was stratified by geographic region (North America, Western Europe, Eastern Europe); 6-minute walk distance ( $\leq 300$  m vs  $> 300$  m); cardiac transplant site vs not; and cardiac resynchronization therapy (CRT) with or without defibrillator recipient (yes/no).

The design incorporated adaptive sample sizing ranging between 400 and 1,000 patients. We used Bayesian predictive probabilities to determine whether the enrolled sample size is sufficient or, alternatively, whether continuing accrual was futile.<sup>8</sup> Scheduled interim analyses occurred at 500 randomized patients and every 100 patients thereafter. If the probability of success with current enrollment and  $\geq 16$ -month follow-up was sufficiently high ( $\geq 0.95$ ), accrual would stop. Beginning with scheduled interim analysis at 400 randomized patients, if the probability of success with maximum sample size was sufficiently low ( $< 0.01$ ), the study would stop for futility. Otherwise, enrollment would continue until the next interim analysis or 1,000 patients were enrolled. Setting 1-sided primary endpoint significance at 0.019 controls the overall type I error at a 1-sided 0.025 level.

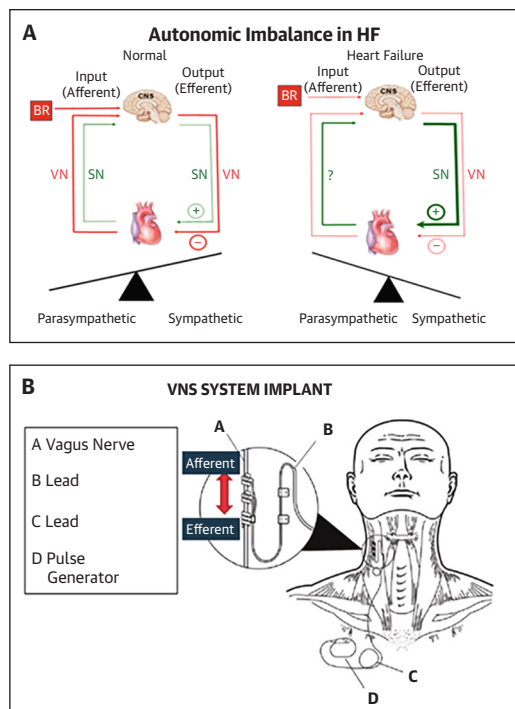
All participants provided written informed consent, and the trial protocol was approved by Institutional Review Boards or Ethics Committees ([Supplemental Table 1](#)).

**PATIENT ELIGIBILITY.** Key inclusion criteria were stable GDMT for  $\geq 4$  weeks, either NYHA functional class III or II, combined with heart failure hospitalization (HFH) within the previous 12 months, left ventricular ejection fraction (LVEF)  $\leq 35\%$ , left ventricular (LV) end-diastolic diameter  $< 8.0$  cm, N-terminal pro-B-type natriuretic peptide (NT-proBNP)

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**CENTRAL ILLUSTRATION** Background and Key Elements of the Trial

| C                                                                                                                                                                                                                                                                                                                                     |     |                                                 |                             |                                 | ANTHEM-HFrEF Pivotal Study |           |                                            |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------|-----------------------------|---------------------------------|----------------------------|-----------|--------------------------------------------|--|
| Plan: Primary outcome—time to CVD or first HFH. Up to 1,000 pts with all patients followed for ≥ 16 mos.                                                                                                                                                                                                                              |     |                                                 |                             |                                 |                            |           |                                            |  |
| Actual: Enrollment curtailed at 532 with median f/u 16.9 mos. Investigators' charge became maintaining transparency, scientific and ethical integrity, and optimizing accurate data output and reporting. Among their actions, they prespecified a single supplemental, exploratory outcome/symptoms endpoint before data unblinding. |     |                                                 |                             |                                 |                            |           |                                            |  |
| D                                                                                                                                                                                                                                                                                                                                     |     |                                                 |                             |                                 |                            |           | Autonomic Engagement: Change in HRV (SDNN) |  |
| Group                                                                                                                                                                                                                                                                                                                                 | n   | LS Mean (95% CI)                                | Treatment v. Control        |                                 |                            |           |                                            |  |
|                                                                                                                                                                                                                                                                                                                                       |     |                                                 | LS mean difference (95% CI) |                                 | P-value (2-sided)          |           |                                            |  |
| Treatment                                                                                                                                                                                                                                                                                                                             | 198 | 3.5 (0.9-5.4)                                   | 7.32 (94.5-10.2)            |                                 | < 0.001                    |           |                                            |  |
| Control                                                                                                                                                                                                                                                                                                                               | 101 | −4.2 (−6.0 to −2.3)                             |                             |                                 |                            |           |                                            |  |
| E                                                                                                                                                                                                                                                                                                                                     |     |                                                 |                             |                                 |                            |           | ANTHEM-HFrEF Primary Endpoints             |  |
| EFFICACY                                                                                                                                                                                                                                                                                                                              |     | Time to CVD or 1 <sup>st</sup> HFH              |                             | HR = 0.84 P = 0.12 (1-sided)    |                            |           |                                            |  |
| SAFETY                                                                                                                                                                                                                                                                                                                                |     | % freedom from implant / device SAE (target 75) |                             | Achieved (CI): 96.7 (94.1-98.3) |                            | P = 0.001 |                                            |  |

Konstam MA, et al. JACC. 2026;■(■):■-■.

(A) Shift in sympathetic and parasympathetic tone in patients with heart failure (HF). (B) Electrode secured onto the right vagus nerve (VN). (C) Original and revised study plans, which handed investigators new challenges and responsibilities. (D) Vagal nerve stimulation (VNS) activity, determined by autonomic engagement. (E) Results of primary efficacy and safety endpoints. CNS = central nervous system; CVD = cardiovascular death; HFH = heart failure hospitalizations; HRV = heart rate variability; ITT = intent-to-treat; LS = least squares; SAE = serious adverse event; SDNN = SD of normal-to-normal (measure of HRV); SN = sympathetic nervous system.

≥800 pg/mL, and a 6MWD of 150 to 450 m limited by HF symptoms. Complete inclusion and exclusion criteria have been published.<sup>8</sup> After study start, the protocol was amended allowing patients with permanent atrial fibrillation, diabetes, or previous CRT to enter.

**TRIAL INTERVENTION.** VNS was titrated over 3 months, either manually at each visit, with programmer and wand, or automatically in small increments several times daily using the system's TitrationAssist. VNS intensity was up-titrated using an 18 second on-time and 66 second off-time, a 2.0- to 3.0-mA current, a 5- to 10-Hz frequency, and a 250-ms pulse width, until autonomic engagement occurred, based on changes in heart rate and heart rate variability (HRV) during signal on-time, seeking sustained autonomic engagement.<sup>2,6,9</sup> A 24-hour electrocardiogram (ECG) was recorded at each visit, with

measurement of heart rate and HRV, to determine whether autonomic engagement persisted.

At maximum neural stimulation intensity of 3.5 mA and electrode separation ≥3.0 cm, there is no detectable interference with either bipolar or unipolar sensing.<sup>10</sup> In clinical practice, electrode separation is maintained ≥15 cm. In vivo, VNS systems are unaffected by repeated high-energy internal or external defibrillation shocks.<sup>11</sup>

**TRIAL ENDPOINTS.** The primary efficacy endpoint was time to first event for cardiovascular (CV) death or first HF hospitalization, as adjudicated by the CEC. A study to evaluate changes in LVEF, 6MWD, and Kansas City Cardiomyopathy Questionnaire (KCCQ) Overall Summary Score (OSS) was embedded within the primary study.<sup>8</sup>

Deaths and hospitalizations were adjudicated by an outcomes group, blinded to randomization

assignment, within the CEC. A separate, unblinded safety group adjudicated implantation- or device-related serious adverse events occurring within the first 90 days postimplant, unanticipated adverse events, and adverse events designated “of special interest” (VNS-associated syncope or ventricular arrhythmias). Blinded assessors conducted the 6-minute walk tests and evaluated NYHA functional class. Informed consent included contingencies for patient withdrawal or loss to follow-up, allowing continued data collection, using a questionnaire regarding safety and health and vital outcomes. Independent core laboratories analyzed NT-proBNP and assessed echocardiography and electrocardiography. The primary safety endpoint was 90-day event-free rate, from all implantation- or system-related serious adverse events.

**STATISTICAL ANALYSIS.** The principal analysis of the primary efficacy endpoint used a log-rank test, with significance set at 1-sided  $P = 0.019$ , and Cox proportional hazards model to estimate HR. The same approach was used for additional analyses evaluating time to CV death and time to HFH. The FDA approved our choice of 1-sided testing at a significance level of 0.025, adjusted for multiplicity, given the hypothesis that the device improves outcomes, compared with control, without intent to test for significant device inferiority. Rather, any trend toward excess morbidity and mortality would preclude approval.<sup>12</sup>

The FDA set the performance goal for the primary safety endpoint at 75% 90-day freedom from procedure- and system-related adverse events. An exact 1-sided binomial test with an overall 1-sided significance level of 0.05 was used.

Between-group change from baseline to the 9-month visit for symptomatic function endpoints 6MWD, KCCQ, LVEF, and NYHA functional class were evaluated using generalized estimation equations (GEE), with a missing at random assumption for missing data. The models incorporated 3-, 6-, and 9-month visits, used an unstructured covariance matrix, and included a covariate for the baseline value. For NYHA functional class, an ordinal logistic regression model was used to compare treatment vs control. The common log OR for the treatment compared with control was estimated with an OR <1 denoting treatment benefit (larger decrease in NYHA functional class from baseline). Model convergence issues necessitated a statistical analysis plan (SAP)-specified independent correlation structure instead of the unstructured correlation structure. Sensitivity analyses for the GEE models’ missing at random

assumptions were performed for each of the 4 metrics. These sensitivity analyses were as follows: 1) last observation carried forward; 2) imputation of zeros for change from baseline for each visit; 3) imputation of mean change in the control group within each visit applied to both treatment arms by visit; and 4) imputation of the smallest change of all individuals within each visit applied to all patients in both treatment groups by visit. The statistical analyses for the several secondary, sensitivity, and exploratory analyses did not include adjustment for multiple testing, so that caution should be deployed in interpreting these results.

The maximum sample size of 1,000 participants was chosen after analysis of multiple simulations based on estimating varying control arm rates of the primary composite (based on historical rates from HF trials) and varying estimated VNS effect sizes. For example, using the maximum sample size of 1,000, minimum patient follow-up of 16 months post-randomization, a global accrual rate of 26 patients/mo (0.29 per site per month) and 10% dropout, the trial had  $\approx 80\%$  power for demonstrating VNS superiority over control when the true HR is 0.70 with a 3-year event rate in the control arm of 43%. The predefined interim analyses to consider sample size reduction (without reducing minimum patient follow-up) for the primary endpoint used accumulating data and explicit algorithm and design parameters. The nominal significance level of 0.019 required to claim superiority was determined by simulating thousands of virtual data sets from multiple sets of plausible assumptions, applying the adaptive algorithm and decision rules to each data set, and ensuring that the probability of erroneously concluding benefit in the null setting is no larger than 0.025 for each set of plausible assumptions.

**ANALYSIS OF 24-HOUR ECG TRACINGS.** Changes from baseline and between-group differences in heart rate and HRV, measured as the SD of normal to normal beat intervals, were analyzed by repeated measures analysis, using a GEE model, in both the intention-to-treat (ITT) and as-treated populations.

**PREMATURE TRIAL DISCONTINUATION AND RESULTING ADDITIONAL STATISTICAL ANALYSES.** Following the sponsor’s decision to terminate enrollment, the executive committee met, while remaining blinded to all trial results, and recognized that early study termination implied reduced sample size and reduced follow-up time, compromising the likelihood of meeting the primary endpoint. The Committee did not alter the prespecified primary analysis; rather it agreed to amend the SAP primarily

by including a single, prespecified (before unblinding and to database lock), exploratory supplemental composite endpoint and statistical test, appropriate for the attenuated sample size and follow-up. (See [Supplemental Appendix](#) containing the SAP amendment. Certain sections of the SAP amendment were redacted at the request of the sponsor to remove confidential information.) This endpoint was a hierarchical composite comprising time to CV death, total HFHs, and KCCQ OSS, evaluated at 9 months, which would be analyzed using the win-ratio statistic and the Finkelstein Schoenfeld test. The Committee also included sensitivity analyses for the primary outcome, addition of 2 patient subgroups—enrollment pre- and post-COVID and NT-proBNP <1,600 pg/mL vs ≥1,600 pg/mL—investigation of possible temporal change, and repeating analyses in an as treated population. No formal adjustments of type I error were made for these exploratory analyses. The FDA reviewed and approved the amended statistical analysis plan before database lock. The study was closed on May 31, 2023, and the study database was locked on July 15, 2023. All personnel involved in study conduct remained blinded prior to database lock.

## RESULTS

Enrollment began on May 23, 2018. There were 84 active study sites in the United States, Austria, Belgium, Bulgaria, Canada, Czech Republic, Germany, Greece, Italy, Norway, Romania, Serbia, and the United Kingdom. A total of 1,065 patients signed informed consent and underwent screening, and 533 patients were randomized. There were 206 patients (38.7%) randomized before the World Health Organization's declaration of the COVID-19 pandemic. One patient (0.2%) was incorrectly randomized after death and was excluded from the ITT population, leaving 358 patients randomized to VNS plus GDMT and 174 patients randomized to GDMT alone, resulting in 532 ITT patients. Median follow-up was 16.9 months (Q1-Q3: 6.8-29.3 months). In total, 16 patients (4.5%) randomized to the treatment group did not undergo VNS implantation. Seven patients (1.2%) were implanted but did not subsequently complete the first titration visit.

There were 43 study discontinuations (8.1%) comprising 39 withdrawals (7.3%) and 4 follow-up losses (0.8%). Withdrawals with no further contact were low in both the treatment (3.1%) and control

**TABLE 1** Baseline Characteristics (ITT Population, n = 532)

|                                  | Treatment<br>(n = 358 [67.3%]) | Control<br>(n = 174 [32.7%]) |
|----------------------------------|--------------------------------|------------------------------|
| Age, y                           | 65.1 ± 9.7                     | 62.4 ± 10.9                  |
| Male                             | 299 (83.5)                     | 145 (83.3)                   |
| Caucasian                        | 332 (92.7)                     | 164 (94.3)                   |
| NYHA functional class            |                                |                              |
| II                               | 56 (15.6)                      | 28 (16.1)                    |
| III                              | 302 (84.4)                     | 145 (83.3)                   |
| Permanent atrial fibrillation    | 71 (19.8)                      | 32 (18.4)                    |
| Hypertension                     | 292 (81.6)                     | 137 (78.7)                   |
| Ischemic etiology                | 222 (62.0)                     | 97 (55.7)                    |
| Heart rate, beats/min            | 72.2 ± 12.3                    | 73.1 ± 13.1                  |
| Blood pressure, mm Hg            |                                |                              |
| Systolic                         | 116.6 ± 15.2                   | 115.8 ± 15.1                 |
| Diastolic                        | 72.5 ± 10.6                    | 74.1 ± 10.3                  |
| eGFR, mL/min/1.73 m <sup>2</sup> | 58.6 ± 24.9                    | 66.0 ± 27.7                  |
| LVEF, %                          | 26.1 ± 6.1                     | 25.9 ± 5.6                   |
| NT-proBNP, pg/mL                 | 3,772.3 ± 4,347.5              | 3,738.0 ± 5,326.65           |
| First 6MWD, m                    | 309.2 ± 82.8                   | 313.8 ± 82.6                 |
| Second 6MWD, m                   | 307.2 ± 81.5                   | 310.7 ± 84.9                 |
| Baseline medications             |                                |                              |
| ACE-ARB-ARNI                     | 303 (84.6)                     | 155 (89.1)                   |
| ARNI                             | 144 (40.1)                     | 66 (37.9)                    |
| Beta-blocker                     | 342 (95.5)                     | 169 (97.1)                   |
| MRA                              | 298 (83.2)                     | 140 (80.5)                   |
| SGLT2 inhibitor                  | 60 (16.7)                      | 20 (11.5)                    |
| Any CRT (CRT or CRT-D)           | 79 (22.1)                      | 36 (20.7)                    |
| Any ICD (CRT-D or ICD)           | 148 (41.3)                     | 74 (42.5)                    |

Values are mean ± SD or n (%).

6MWD = 6-minute walk distance; ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; CRT = cardiac resynchronization therapy; CRT-D = cardiac resynchronization therapy with defibrillator; eGFR = estimated glomerular filtration rate; ICD = implantable cardioverter-defibrillator; ITT = intention-to-treat; LVEF = left ventricular ejection fraction.

group (4.6%). Fewer patients in the treatment group (2.2%) than in the control group (4.6%) underwent LV assist device implantation or cardiac transplantation with subsequent censoring.

**BASELINE CHARACTERISTICS.** [Table 1](#) summarizes patient demographics and baseline treatment by treatment group. The study started before angiotensin receptor/neprilysin inhibitor received first-line recommendation and before sodium-glucose cotransporter 2 inhibitors received regulatory approval. Baseline characteristics and treatments were balanced between treatment groups. More patients were administered aldosterone antagonists and fewer CRTs and ICDs were implanted in patients from Eastern Europe. Comparison of this study population to those of 4 other pivotal studies in international populations leading to U.S. product

**TABLE 2** Summary of Key Overall Study Outcomes in the ITT Population (n = 532)

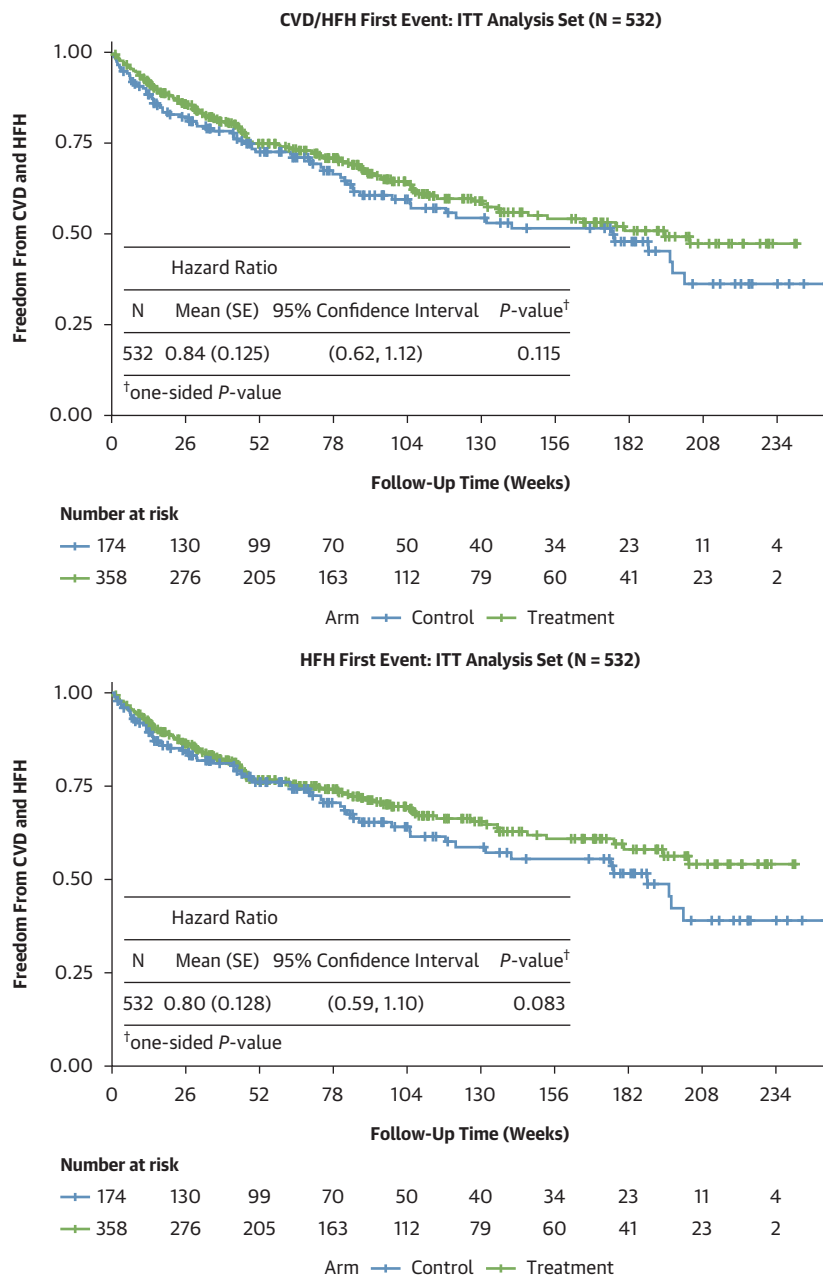
|                                                                                   | Outcome | 95% CI         | P Value | Endpoint Met (Yes/No) <sup>a</sup> |
|-----------------------------------------------------------------------------------|---------|----------------|---------|------------------------------------|
| Primary study of morbidity and mortality                                          |         |                |         |                                    |
| Primary efficacy endpoint: time to first event: CV death or HF hospitalization—HR | 0.84    | 0.62-1.12      | 0.115   | No                                 |
| Time to CV death—HR                                                               | 0.90    | 0.58-1.42      | 0.332   |                                    |
| Time to first HF hospitalization—HR                                               | 0.80    | 0.59-1.10      | 0.083   |                                    |
| Total HF hospitalizations                                                         |         |                |         |                                    |
| Control                                                                           | 290     |                |         |                                    |
| VNS                                                                               | 186     |                |         |                                    |
| Rates of HFHs per patient-y                                                       |         |                |         |                                    |
| Control                                                                           | 0.43    | 0.36-0.50      |         |                                    |
| VNS                                                                               | 0.38    | 0.33-0.43      |         |                                    |
| Primary safety endpoint                                                           |         |                |         |                                    |
| % freedom from implant and device SAEs within 90 d—comparison to target 0.75      | 96.7    | 94.1-98.3      | <0.001  | Yes                                |
| Hierarchical composite outcome and health status (WR)                             |         |                |         |                                    |
| Overall composite outcome - WR overall                                            | 1.25    | 1.00-1.55      | 0.023   | Yes                                |
| (W/L% of overall pairings)                                                        | (85.4)  |                |         |                                    |
| CV death (time to event)                                                          |         |                |         |                                    |
| WR contribution                                                                   | 1.22    |                |         |                                    |
| (W/L% of overall pairings)                                                        | (19.3)  |                |         |                                    |
| HF hospitalizations                                                               |         |                |         |                                    |
| WR contribution                                                                   | 1.16    |                |         |                                    |
| (W/L% of overall pairings)                                                        | (26.9)  |                |         |                                    |
| KCCQ overall score                                                                |         |                |         |                                    |
| WR contribution                                                                   | 1.32    |                |         |                                    |
| (W/L% of overall pairings)                                                        | (39.2)  |                |         |                                    |
| Residual ties, %                                                                  | 14.6    |                |         |                                    |
| Symptom and function endpoints                                                    |         |                |         |                                    |
| KCCQ Overall Summary Score, points                                                |         |                |         |                                    |
| Mean between group difference in change from baseline                             | 7.3     | 3.02-11.61     | <0.001  | Yes                                |
| Missing at random analysis                                                        |         |                |         |                                    |
| 6MWD, m                                                                           |         |                |         |                                    |
| Mean between group difference in change from baseline                             |         |                |         |                                    |
| Missing at random analysis                                                        | 14.8    | -1.93 to 31.55 | 0.041   | Trend                              |
| Imputation—last observation carried forward                                       | 16.2    | 1.03 to 31.29  | 0.018   | Yes                                |
| LVEF, %                                                                           |         |                |         |                                    |
| Mean between group difference in change from baseline                             | -0.45   | -1.65 to 0.76  | 0.765   | No                                 |
| Missing at random analysis                                                        |         |                |         |                                    |
| NYHA functional class                                                             |         |                |         |                                    |
| Improvement in NYHA <sup>b</sup>                                                  | 0.61    | 0.38-0.96      | 0.017   | Yes                                |

P values are 1-sided. <sup>a</sup>Endpoint achieved nominal statistical significance (1-sided P value ≤ 0.025). <sup>b</sup>Odds ratio, treatment vs control (value <1 denotes treatment benefit). HF = heart failure; KCCQ = Kansas City Cardiomyopathy Questionnaire; SAE = serious adverse event; VNS = vagal nerve stimulation; W/L = win/loss; WR = win ratio; other abbreviations as in Table 1.

approvals shows that the present population was sicker than those in the other trials, based on NYHA functional class and LVEF. However, adherence to GDMT appeared equivalent to the comparator trials (Supplemental Table 2).<sup>13-16</sup>

**EFFICACY RESULTS.** The Central Illustration summarizes the trial's key events and findings. Table 2 shows comparisons of the treatment and control groups for the principal trial outcomes. The primary

efficacy endpoint was not met. Figure 1 provides the Kaplan-Meier curves for the primary composite efficacy endpoint and the HFH component within the ITT population (n = 532). The point estimates of the HR for the primary efficacy endpoint, time to CV death, and time to first HFH were directionally favorable for the treatment group but did not reach statistical significance (Table 2). The primary efficacy endpoint occurred in 34.6% in the treatment group and 40.2% in the control group. The total

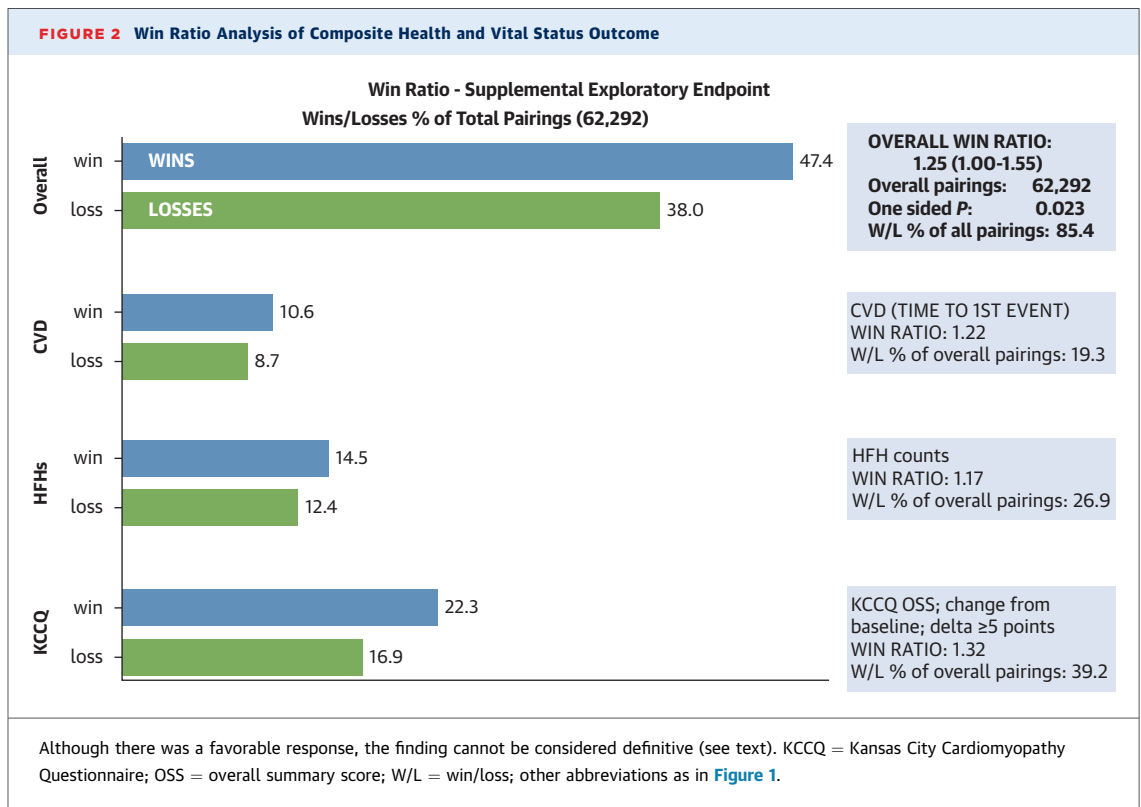
**FIGURE 1** Kaplan-Meier Curve

(A) Primary endpoint of cardiovascular death (CVD) and heart failure hospitalization (HFH) (intention-to-treat [ITT] population). (B) HFH (time to event; ITT population).

hospitalizations per patient-year was also not significantly different, amounting to 0.38 in the treatment group (95% CI: 0.33-0.43) and 0.43 in the control group (95% CI: 0.36-0.50). The treatment effect on the primary efficacy outcome was

directionally consistent in favor of the treatment group across prespecified subgroups in the ITT population, with no subgroup pair displaying significant treatment-by-subgroup interaction (Supplemental Figure 1).





There were 195 primary endpoint events, with a median exposure of 67 weeks. Applying the event rate to a maximum sample size of 1,000 and minimum follow-up  $\geq 16$  months would generate 450 to 550 events resulting in a retained information fraction of 0.35 to 0.43. Conditional on the observed data at the premature sample size of 532 patients, the Bayesian predictive probability of success at the maximum sample size was 0.31. Based on simulations, the largest HR obtaining statistical significance at sample size =1,000 was approximately 0.85, and the smallest HR that failed to achieve significance was approximately 0.80. Hence, the observed HR of 0.84 with the premature sample size of 532 is consistent with an effect that would be “borderline” at a sample size of 1,000, with either a positive or neutral primary endpoint.

Data for the treatment vs control at 9 months in the embedded study of the ITT population are reported in [Table 2](#). There was nominally significant improvement in KCCQ OSS for treatment vs control. There was a favorable trend in 6MWD improvement based upon a missing at random analysis. There was no significant change in LVEF for treatment vs control. The OR for NYHA functional class improvement was nominally significant for treatment vs control.

[Figure 2](#) summarizes the prespecified, supplemental, exploratory win ratio composite in the ITT population. The overall Finkelstein-Schoenfeld test was nominally significant ( $P$  [1-sided] = 0.023) with a win ratio of 1.25 (95% CI: 1.00-1.55). There was a larger number of pairwise wins in the treatment group relative to control group for each of the 3 components: time to CV death, total HFHs, and KCCQ-OSS. The win ratio findings were consistent across key subgroups ([Supplemental Figure 2](#)). Although promising, the exploratory win ratio cannot be considered conclusive, given a neutral primary endpoint, concerns about multiplicity, potential bias in the subjective KCCQ assessment, and marginal statistical significance.

Patient experience with the device was generally positive. Among patients receiving VNS, surveyed at 9 months (265 patients, 79.1% of treatment group), 63.4% reported that they were likely to recommend VITARIA to others with a similar health condition, whereas 12.5% reported being unlikely to recommend VITARIA.

**AUTONOMIC ENGAGEMENT.** VNS treatment achieved sustained autonomic modulation. Overall and at most visits, 24-hour ECGs showed increased HRV,



**TABLE 3** Summary of Patients With Reported SAEs by Treatment Group—Safety Population (n = 516)

|                                                    | Treatment<br>(n = 342) | Control<br>(n = 174) |
|----------------------------------------------------|------------------------|----------------------|
| Patients with SAEs                                 | 203 (59.4)             | 103 (59.2)           |
| Patients with SAEs leading to                      |                        |                      |
| Treatment interruption                             | 7 (2.0)                | 1 (0.6)              |
| Permanent treatment discontinuation                | 11 (3.2)               | NA                   |
| Complete (permanent) study withdrawal <sup>a</sup> | 10 (2.9)               | 5 (2.9)              |
| Death                                              | 82 (24.0)              | 46 (26.4)            |

Values are n (%). <sup>a</sup>Complete withdrawal with specification of no further contact by study personnel.

SAE = serious adverse event.

measured as SD of normal to normal, in the treatment group, compared to baseline and to control, in both the ITT and as-treated populations (Supplemental Tables 3A and 3B). There were no consistent changes in heart rate.

**SAFETY RESULTS.** The primary safety endpoint was met with 96.7% (95% CI: 94.1-98.3) freedom from implantation- or system-related serious adverse events within the first 90 days postimplant (Table 2), in comparison with the prespecified performance goal of 75% ( $P < 0.001$ ). Overall safety results are reported in Tables 3 to 5. Overall, serious adverse events occurred in a comparable proportion of patients in the treatment (59.4%) and control groups (59.2%) (Table 3). These events led to death in similar

proportions of patients (24.0% vs 26.4%) and to a low and comparable proportion of permanent study withdrawals in the treatment (2.9%) and control groups (2.9%). Only 2% of device patients with serious adverse events required action taken with VNS treatment. Serious adverse device events were infrequent with the majority requiring no treatment change (Table 4).

Treatment-emergent adverse events occurring with a frequency  $\geq 2\%$  and a significantly greater proportion in the treatment group ranged from 0.6% to 20.2% (Table 5) and were nonserious and comparable in nature and frequency to AEs reported in patients receiving treatment by an approved VNS system for epilepsy or depression.<sup>17</sup> They were mitigated by a brief magnet application over the pulse generator, or programming to adjust VNS intensity.

There were 3 patients with adjudicated device-related serious adverse events of special interest comprising atrioventricular block in 1 patient (0.3%), ventricular tachycardia in 1 patient (0.3%), and presyncope in 1 patient (0.3%). These led to treatment interruption in 1 patient (0.3%), treatment discontinuation in 2 patients ( $< 1.0\%$ ) and no death within the safety population.

## DISCUSSION

Once the trial was prematurely terminated, the investigators understood that their charge was to salvage as much as possible from the trial, while preserving transparency and scientific rigor. At the outset, the trial's purpose was to build upon prior evidence and prove that VNS could safely and effectively yield sustained autonomic engagement, which would improve short- and long-term symptoms and outcomes. The latter goal was thwarted by the limited population size and follow-up duration, precluding any opportunity to hit the primary endpoint. The investigators engaged in a legitimate effort, emphasizing transparency, and prespecifying analytic adaptation before unblinding. Their goal was to salvage the scientific utility of existing data, understanding that they could document feasibility, safety, and autonomic engagement, but that the clinical efficacy findings would not be definitive.

The study goals were justified on the basis of advances in selecting and titrating VNS signal characteristics<sup>5</sup> and on encouraging pilot data in patients with severe HF.<sup>6</sup> Unique features of the protocol included Bayesian adaptive sample sizing and an opportunity for early success based on a combination of meeting the safety goal and demonstrating symptom- and function-related efficacy.<sup>8</sup>

**TABLE 4** CEC-Adjudicated SAEs in the Safety Population Treatment Group (n = 342)

|                                                   |          |
|---------------------------------------------------|----------|
| Patients with adjudicated SAEs <sup>a</sup>       | 16 (4.7) |
| Patients with SAEs                                |          |
| Implant-related                                   | 9 (2.6)  |
| Stimulation-related                               | 5 (1.5)  |
| Other device-related                              | 3 (0.9)  |
| Patients with SAEs leading to:                    |          |
| Treatment interruption                            | 1 (0.3)  |
| Treatment discontinuation                         | 2 (0.6)  |
| Complete withdrawal                               | 0        |
| Death                                             | 0        |
| Action taken with VNS treatment                   |          |
| None required                                     | 13 (3.8) |
| Temporary magnet application over pulse generator | 1 (0.3)  |
| Changed device parameters                         | 0        |
| Stimulation stopped (programmer)                  | 2 (0.6)  |
| Device explantation                               | 1 (0.3)  |

Values are n (%). <sup>a</sup>Patients may have had  $>1$  SAE.

CEC = clinical events committee; SAE = serious adverse device-related event; VNS = vagal nerve stimulation.

**TABLE 5** TEAEs With a Frequency  $\geq 2\%$  of Patients in the Treatment or Control Group With Statistically Significant Differences Between Groups (2-Sided  $P < 0.05$ ) (Safety Population, n = 516)

| Preferred Term               | Treatment (n = 342) | Control (n = 174) | P Value 2-Sided |
|------------------------------|---------------------|-------------------|-----------------|
| Number of patients with TEAE | 303 (88.6)          | 128 (73.6)        | <0.001          |
| Ear pain                     | 20 (5.8)            | 1 (0.06)          | 0.004           |
| Dysphagia                    | 13 (3.8)            | 0                 | 0.009           |
| Nausea                       | 19 (5.6)            | 0                 | 0.002           |
| Toothache                    | 16 (4.7)            | 0                 | 0.004           |
| Medical device pain          | 13 (3.8)            | 1 (0.6)           | 0.033           |
| Hypokalemia                  | 6 (1.8)             | 9 (5.2)           | 0.029           |
| Hyponatremia                 | 2 (0.6)             | 5 (2.9)           | 0.034           |
| Musculoskeletal discomfort   | 13 (3.8)            | 0                 | 0.009           |
| Neck pain                    | 45 (13.2)           | 0                 | <0.001          |
| Pain in jaw                  | 33 (9.6)            | 0                 | <0.001          |
| Headache                     | 48 (14.0)           | 0                 | <0.001          |
| Hypoesthesia                 | 10 (2.9)            | 0                 | 0.023           |
| Paresthesia                  | 29 (8.5)            | 1 (0.6)           | <0.001          |
| Tremor                       | 22 (6.4)            | 0                 | <0.001          |
| Cough                        | 60 (17.5)           | 4 (2.3)           | <0.001          |
| Dysphonia                    | 69 (20.2)           | 0                 | <0.001          |
| Oropharyngeal pain           | 16 (4.7)            | 1 (0.6)           | 0.014           |

Values are n (%).  
TEAE = treatment-emergent adverse event.

The executive committee ensured adherence to the original statistical analysis plan and amended that plan, prespecifying a single supplementary exploratory win ratio composite, defined without knowledge of study results, incorporating outcomes and symptoms, in accordance with the FDA's Guidance on Endpoints in Clinical Trials.<sup>18</sup> Upon data unblinding and analysis, it was learned that the primary endpoint trended favorable to treatment but, as expected, failed to reach statistical significance. There were favorable trends for time to first HFH and total HFH rate, and nominal significance or favorable trends for symptom and function endpoints, consistent with findings of prior studies. It is promising that the differences of all of these endpoints rest in the same, favorable direction, including the 3 win-ratio components. However, it is important to recognize that none of these findings can be considered definitive. None can offset neutrality of the primary endpoint, particularly in the absence of multiplicity adjustment. In addition, in an unblinded trial, any positivity in a subjective endpoint (eg, KCCQ and NYHA functional class) must be questioned in view of the likelihood of bias. Each of these caveats applies to the exploratory mortality-hospitalization-KCCQ win ratio, with the addition of the marginal nature of its  $P$  value.

What, then, can be concluded by the trial's conduct and findings? Beyond documenting feasibility and safety of VNS, maintaining sustained autonomic engagement, and generating important hypothesis-generating findings, it displays the actions of the executive committee, as a model for future investigators faced with early trial termination. Investigator actions included maintaining trial integrity and adhering to the original analytic plan. The supplemental exploratory endpoint proved informative yet inconclusive, but its blinded pre-specification represents a future consideration to transparently extract additional useful information in the setting of a reduced sample size. These actions represent key efforts to derive the greatest possible utility from a truncated trial, while adhering to ethical and scientific standards. It was essential to recognize the mandate to draw out the maximum amount of scientific knowledge out of respect for the patients who consented to participation in a trial that did not reach its planned conclusion.

There are several signals that might point toward strengthening any future investigation:

1. Greater credibility of patient reported outcomes calls for exploring incorporating treatment blinding (eg, device implantation without activation) at least for a period.
2. Although there were no significant treatment-by-subgroup interactions, and over-interpretation of subgroups is hazardous, there are signals across several subgroups that might be further explored, particularly those based on NYHA functional class, heart rate, and geographic region ([Supplemental Figure 1](#)), and point toward possible population enrichment with patients more likely to respond.

The primary safety endpoint was met with the composite substantially exceeding the FDA's performance goal. There were no limiting patient safety concerns or device malfunctions. The safety data are consistent with the large non-CV experience with VNS for epilepsy or depression treatment.<sup>17</sup> The 96.7% (95% CI: 94.1%-98.3%) freedom from implantation or system-related serious adverse events was similar to the 97% (95% CI: 93%-100%) freedom from such events for baroreflex stimulation<sup>19</sup> and to a corresponding figure of 89.7% (95% CI: 79.9%-95.8%) for cardiac contractility modulation.<sup>20</sup>

Prior trials have not shown consistent effects of VNS in patients with HFrEF. NECTAR-HF (Neural Cardiac Therapy for Heart Failure Study), in 96 patients, did not show improvement in LV end-systolic dimension (the primary efficacy endpoint) or other echocardiographic measures, although there was

improvement in the more clinically relevant health-related quality of life.<sup>3</sup> An analysis of trial data indicated that, in most patients, no significant increase was observed in HRV, a marker of physiological autonomic modulation. The INOVATE-HF (Increase of Vagal Tone in Heart Failure) study enrolled 707 patients with chronic HF, NYHA functional class III symptoms and ejection fraction  $\leq 40\%$ .<sup>4</sup> VNS did not improve the rate of CV death or HF events (the primary endpoint) or benefit LV remodeling. Favorable signals were seen in some markers of symptoms or function. In ANTHEM-HFrEF, the persistent increases in HRV likely ensures that the neutrality of the primary endpoint is not a function of inadequate autonomic engagement and allows for future exploration of the relationship between this indicator and clinical outcomes. As shown in [Supplemental Table 2](#), the ANTHEM-HFrEF population appears sicker than in several recent trials, but patients appeared well managed.

**STUDY LIMITATIONS.** The limitations of these findings are clear and predominantly stem from early trial termination. The analytic information available was estimated to be 35% to 43% of what was expected had the trial been completed. Efficacy analyses based on the existing data set remained underpowered and inconclusive caused by neutrality of the primary endpoint, subjectivity in the absence of a sham control, and lack of statistical adjustment for multiplicity.

## CONCLUSIONS

The primary efficacy endpoint was neutral, an expected finding because of the sponsor's premature halting of the trial, causing it to be underpowered. Nevertheless, there are important conclusions and lessons to be drawn. First, the findings demonstrate feasibility of the trial and safety of the device. Second, unlike earlier studies, here VNS was shown to achieve and sustain autonomic engagement. Third, there were favorable trends in the primary efficacy outcome and HFH, and there were signals for improved patient reported outcomes. These findings were echoed by the supplemental win-ratio composite and each of its components, and consistent with findings from prior VNS studies. These results should be considered hypothesis-generating, requiring future confirmatory investigation. The subjective endpoints in this unblinded investigation should be interpreted with caution and highlight the importance of incorporating treatment blinding wherever feasible. Importantly, concerning study conduct, the approach taken after premature trial

termination underscores the investigators' need to fulfill their scientific and ethical responsibilities to preserve trial integrity and transparency, while maximizing the knowledge to be gained, particularly out of respect for the patients who volunteered to participate. In the absence of a definitive efficacy finding, the trial's importance lies in the conclusions that can be reached, the generation of hypotheses to fuel future investigation, and providing a model for preserving scientific rigor and ethical integrity, under these circumstances.

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Throughout the paper, we refer to actions taken by "the investigators." That term applies to every author on this paper, regardless of employment by an academic medical center or by industry. In particular, those listed who are employed by industry were all members of the study team, and each contributed importantly to the scientific effort, adhering to the highest principles of scientific conduct. These individuals played no role in the decision to terminate the trial and remained blinded until data lock. They played an invaluable role in developing and executing our response following trial termination and in achieving FDA approval of our statistical analysis plan addendum.

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## PERSPECTIVES

ANTHEM-HFrEF explored the clinical value of VNS in patients with moderately severe HF and reduced LVEF. The study was prematurely stopped. The primary efficacy endpoint of time to CV death or HFH was not met in the context of limited sample size and follow-up. Favorable trends in morbidity, symptoms, and function, including findings of a hierarchical win ratio analysis must be considered exploratory, yet hypothesis-generating. Although the primary endpoint was neutral, ANTHEM-HFrEF provides important insights into feasibility, safety, and trial conduct following premature termination. Alternate means of neurohormonal modulation exist, and clinical applicability of these and other such interventions will become clearer through ongoing and future investigation.

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**KEY WORDS** autonomic nervous system, autonomic regulation therapy, heart failure, neuromodulation, reduced ejection fraction, vagus nerve stimulation

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**APPENDIX** For supplemental tables and figures, trial protocol, statistical analysis, and a full list of investigators and committee members, please see the online version of this paper.